
Assignment 5 Branch and Jump Instructions

The fifth assignment extends the implemented instruction set to include the B-type and J-type instructions from the RV32I ISA.

Structure of the Project

This task uses your implementation from task 4 (5-stage pipeline with forwarding unit) as a starting point. Make sure that your core is working correctly before starting with this assignment.

Task 5.1 Preparation

Before you start with the implementation, please answer the following questions:

1. Where can you find the encoding of the branch and jump instructions in RISC-V?
2. What new instruction format do you need to support for the branch and jump instructions?
3. What stages of your pipeline do you need to modify to support the branch and jump instructions?
4. What new types of hazards can arise with branch and jump instructions, and how can they be resolved?

Task 5.2 Implementation

Add all branch and jump instructions from the RV32I ISA subset to your pipelined 5-stage 32-bit RISC-V processor core. This includes the following instructions: beq, bne, blt, bge, bltu, bgeu, jal, and jalr. Pay attention to enabling the core to handle control hazards that arise from branch and jump instructions correctly. For the beginning, you can use a simple static branch prediction scheme that always assumes conditional branches as not taken. Unconditional jumps (jal, jalr) should always be taken. Implement the necessary control logic to flush the pipeline when a conditional branch turns out to be taken.

Task 5.3 Test your Implementation

Check whether your design runs and produces correct results by adapting the test cases you created in the previous assignments. Make sure to add test cases that specifically test different interactions of branch and jump instructions. The test bench is located in `PipelinedRISCV32I.tb.scala`, test cases need to be added in HEX assembly to the "Binary File" in the "programs" folder.

A [RISC-V encoder](#) and a [RISC-V interpreter](#) might be helpful for you to achieve this task.